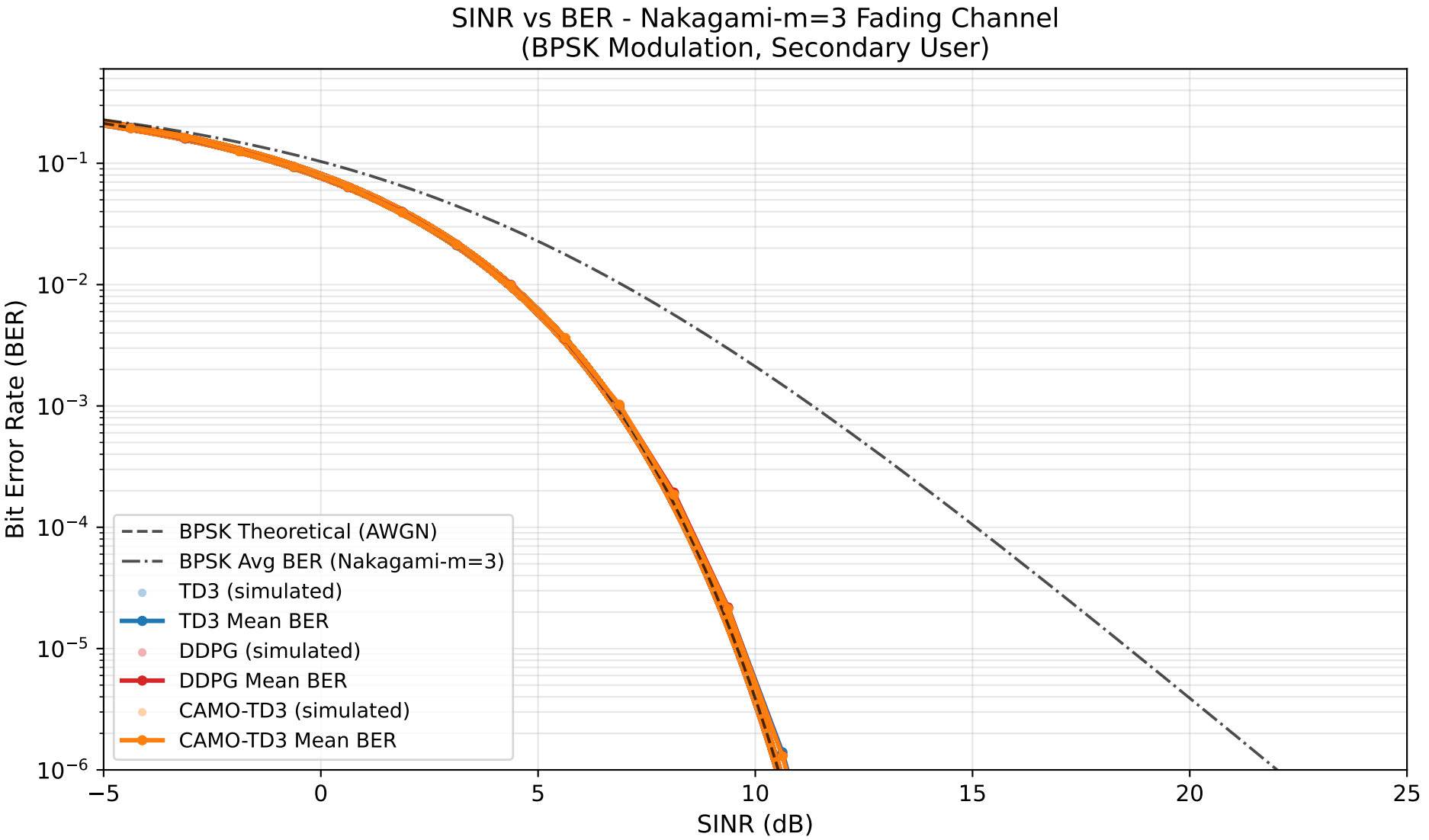
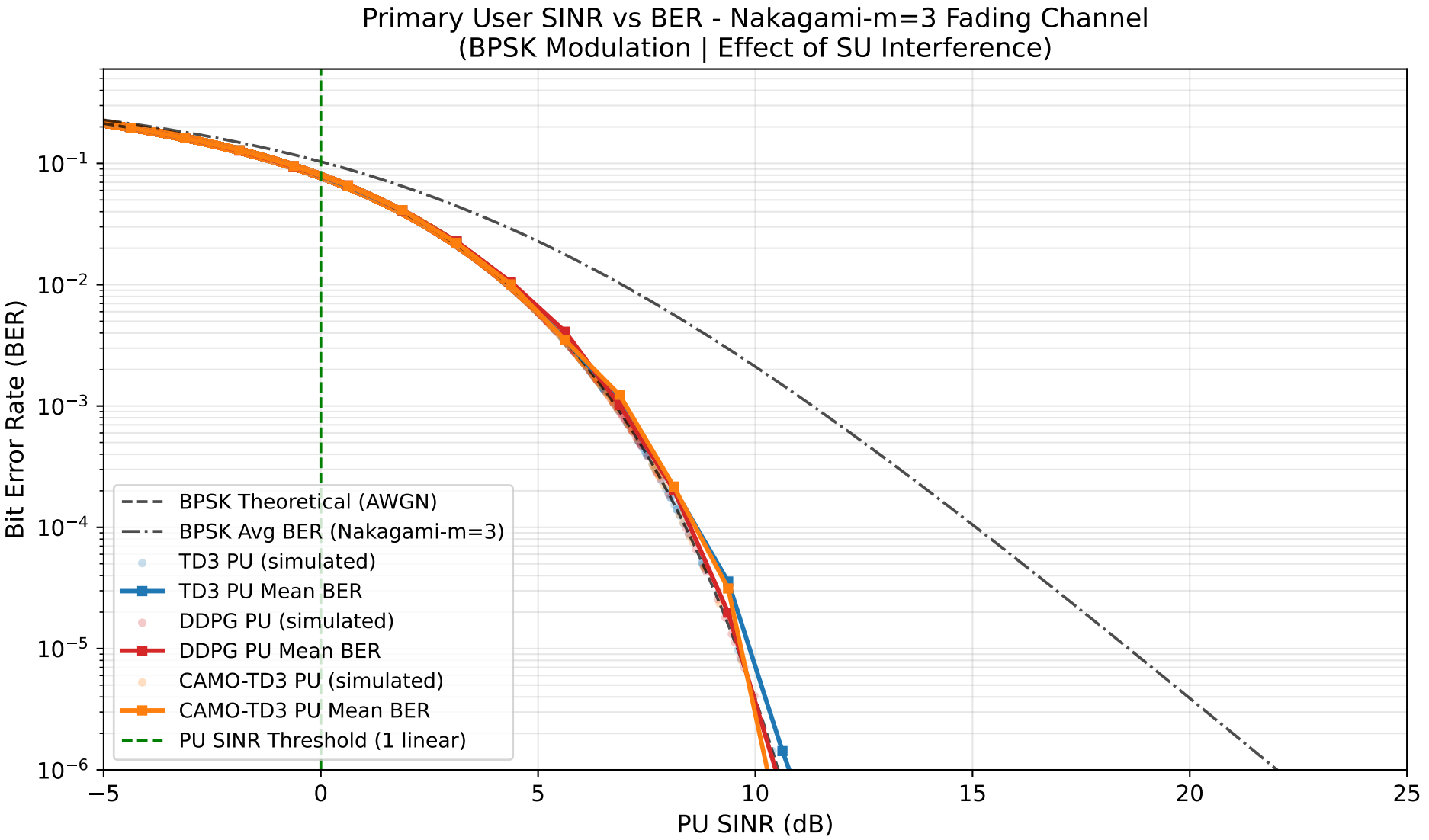
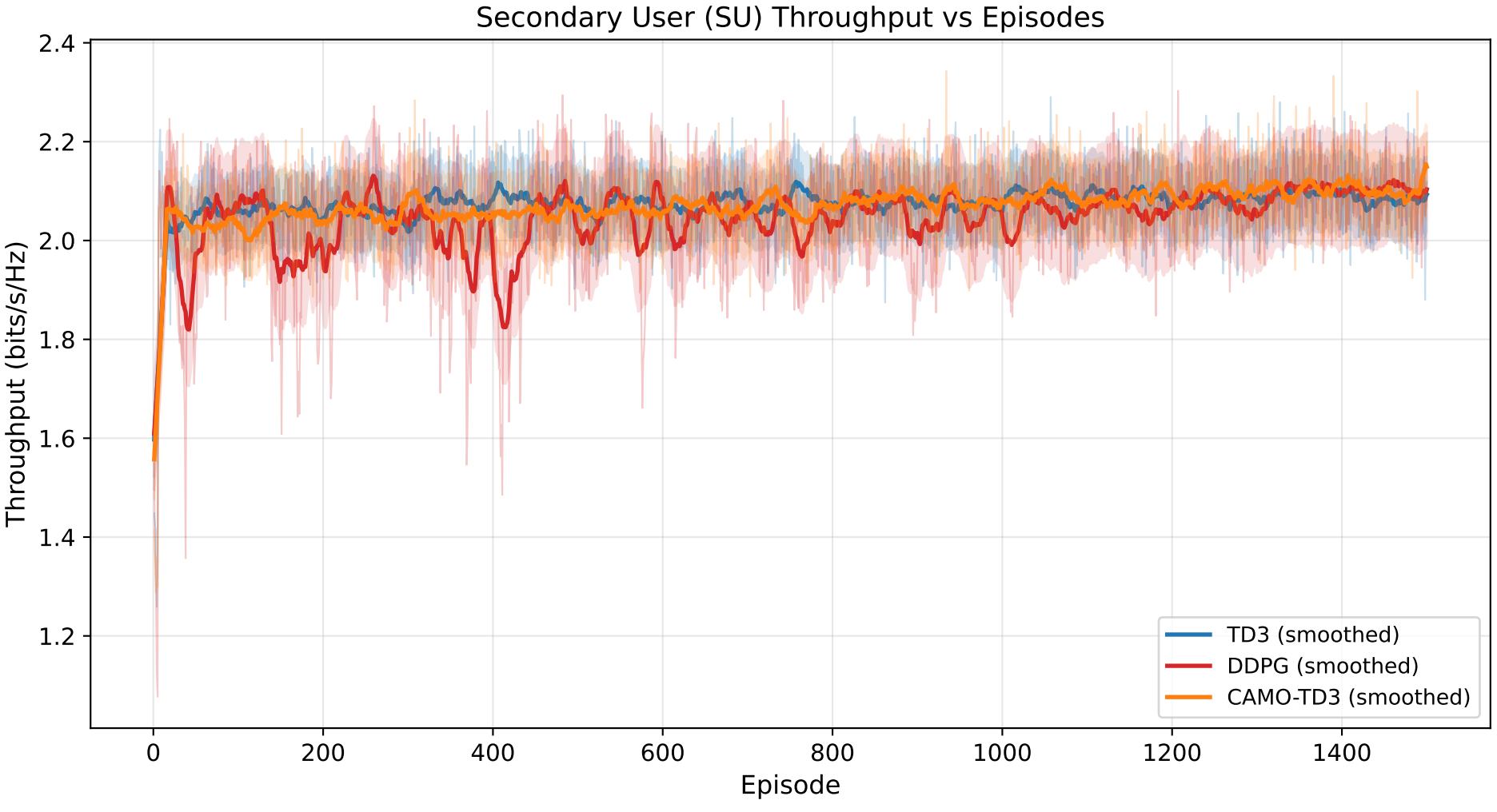
CRN Power Control: TD3 vs DDPG vs CAMO-TD3
Nakagami-m Fading Channel Performance Report

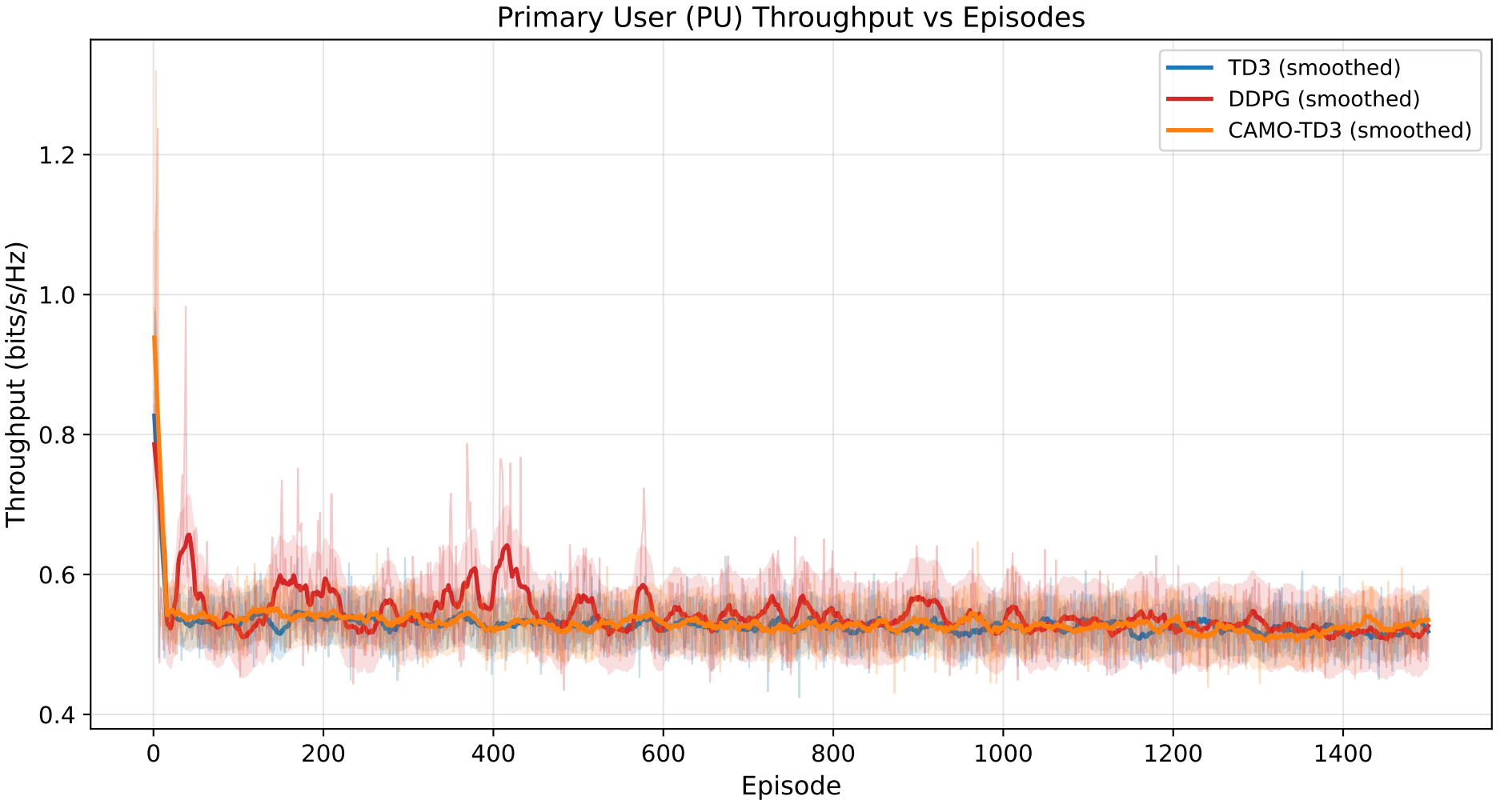
Nakagami-m = 3.0 | Episodes = 1500 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

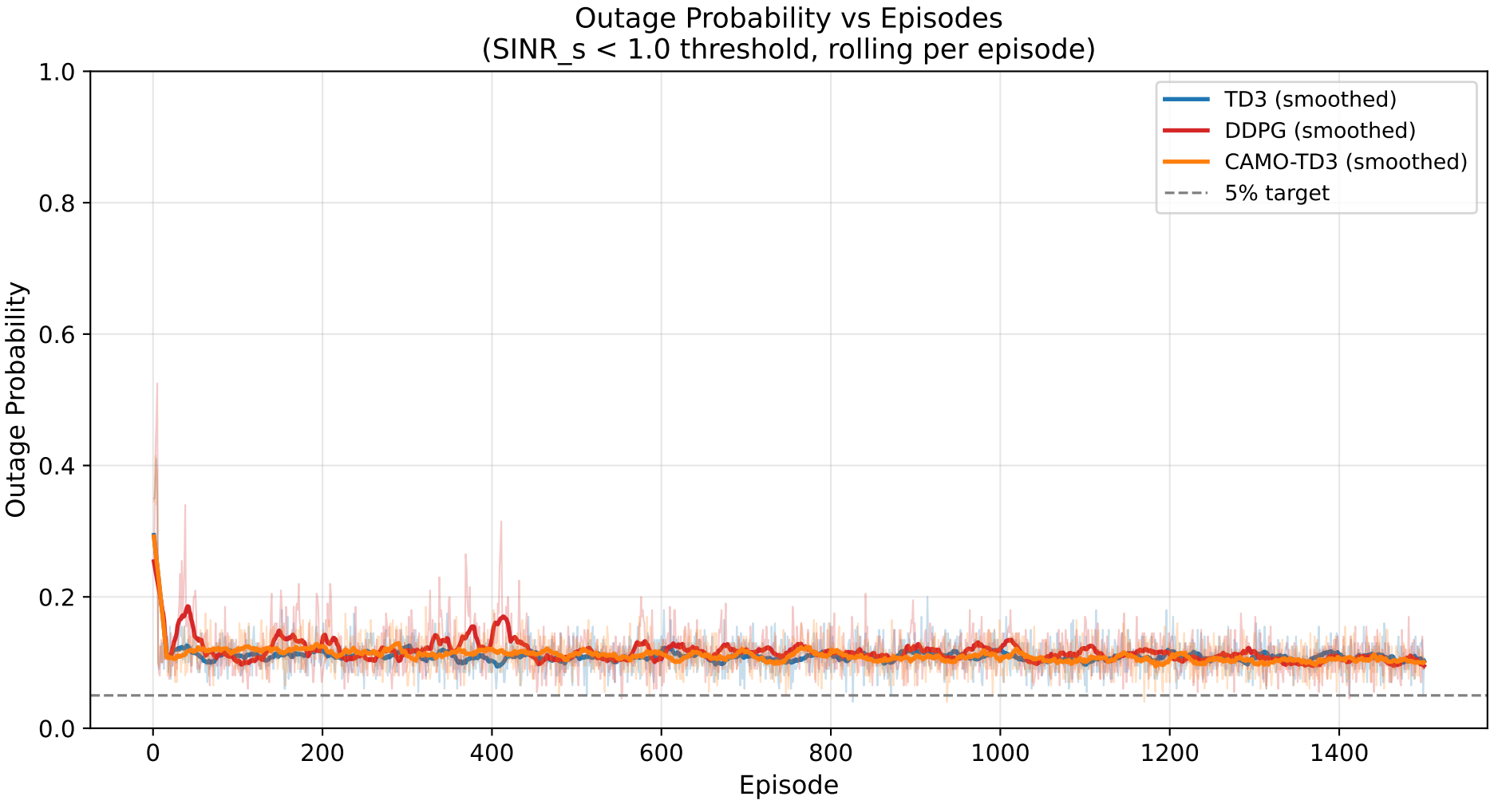
Metric	TD3	DDPG	CAMO-TD3	Winner
Avg Reward (last 100 ep	241.3344	245.4553	247.5641	**CAMO-TD3**
SU Throughput (bits/s/Hz	2.0814	2.1034	2.1002	**DDPG**
PU Throughput (bits/s/Hz	0.5186	0.5161	0.5292	**CAMO-TD3**
Outage Probability	0.1065	0.1041	0.1043	**DDPG**
Average BER	0.027016	0.026309	0.026376	**DDPG**
Training Time (s)	5486.7	5213.8	44386.5	-

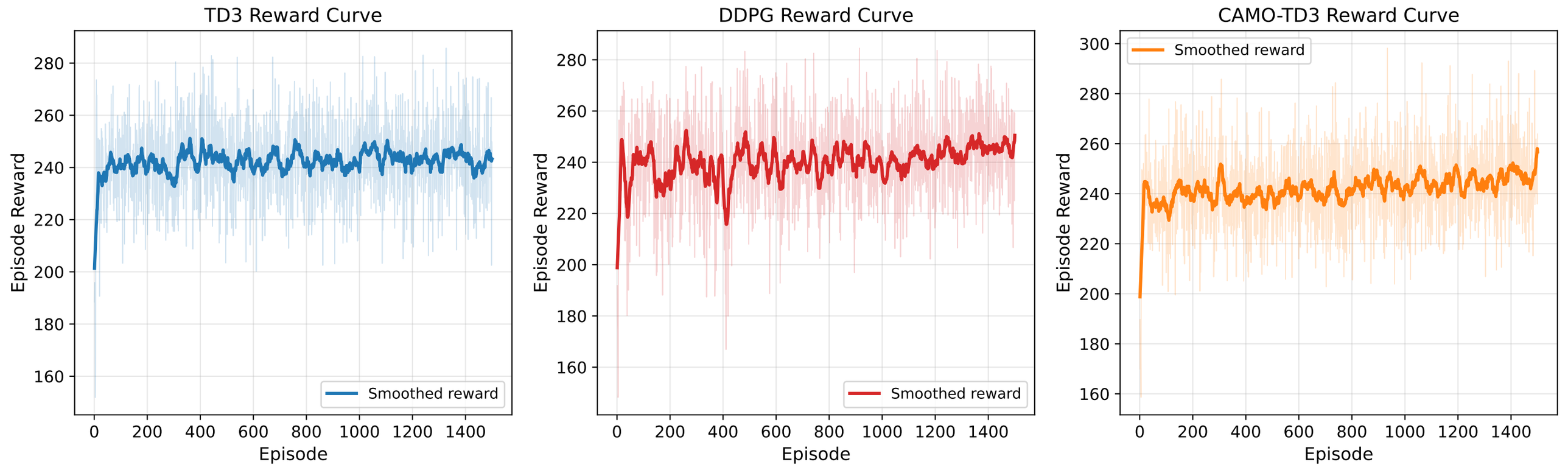

SINR vs BER - Nakagami-m=3 Fading Channel
(BPSK Modulation, Secondary User)
Bit Error Rate (BER)
SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 (simulated)
TD3 Mean BER
DDPG (simulated)
DDPG Mean BER
CAMO-TD3 (simulated)
CAMO-TD3 Mean BER

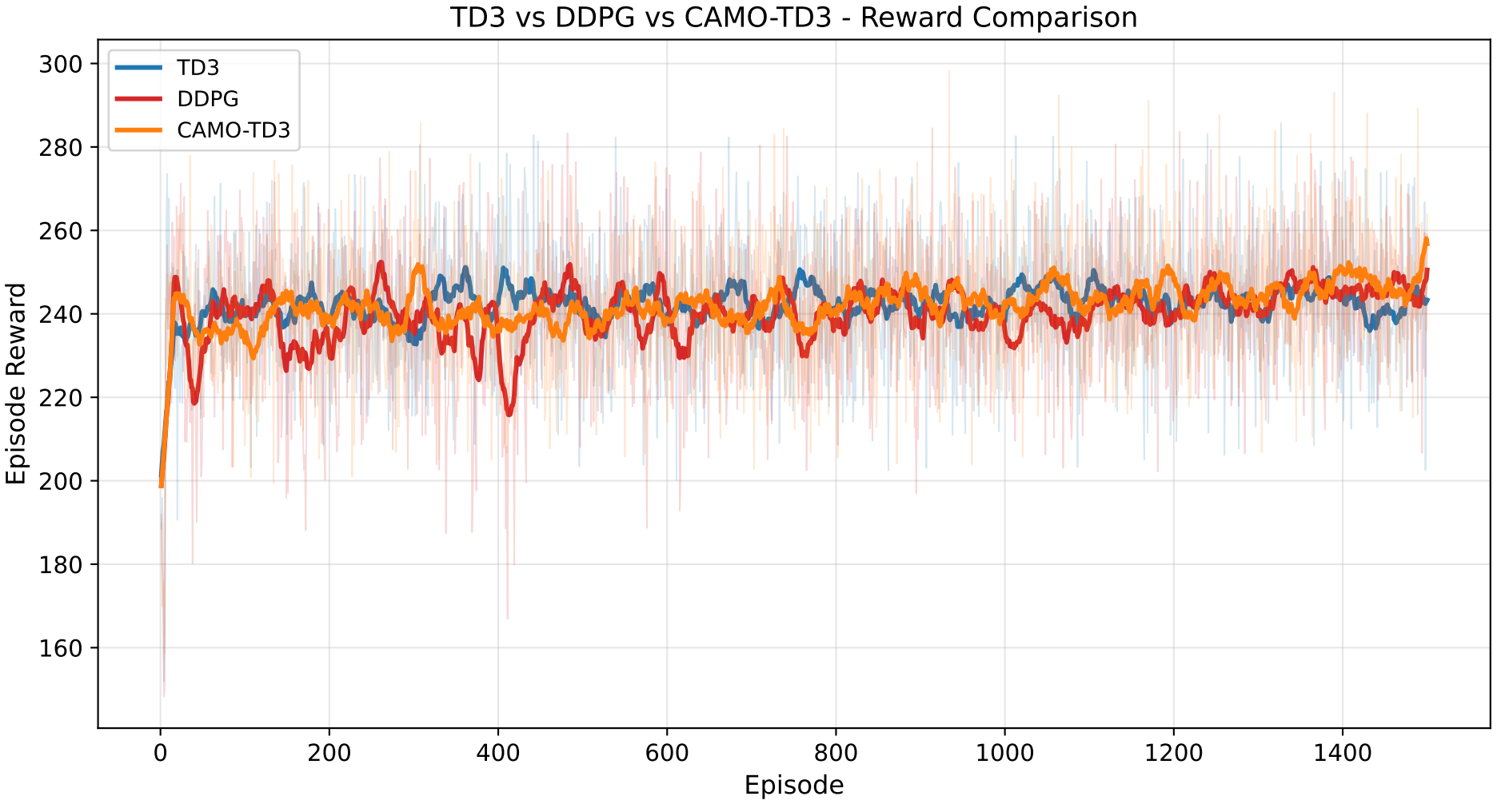

Primary User SINR vs BER - Nakagami-m=3 Fading Channel
(BPSK Modulation | Effect of SU Interference)
Bit Error Rate (BER)
PU SINR (dB)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
−5
0
5
10
15
20
25
BPSK Theoretical (AWGN)
BPSK Avg BER (Nakagami-m=3)
TD3 PU (simulated)
TD3 PU Mean BER
DDPG PU (simulated)
DDPG PU Mean BER
CAMO-TD3 PU (simulated)
CAMO-TD3 PU Mean BER
PU SINR Threshold (1 linear)


Secondary User (SU) Throughput vs Episodes
Throughput (bits/s/Hz)
Episode
2.4
2.2
2.0
1.8
1.6
1.4
1.2
0
200
400
600
800
1000
1200
1400
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)


Primary User (PU) Throughput vs Episodes
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)
Throughput (bits/s/Hz)
1.2
1.0
0.8
0.6
0.4
0
200
400
600
800
1000
1200
1400
Episode


Outage Probability vs Episodes
(SINR_s < 1.0 threshold, rolling per episode)
TD3 (smoothed)
DDPG (smoothed)
CAMO-TD3 (smoothed)
5% target
Outage Probability
Episode
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
1400


Episode Reward Curves - TD3 vs DDPG vs CAMO-TD3
TD3 Reward Curve
DDPG Reward Curve
CAMO-TD3 Reward Curve
Smoothed reward
Smoothed reward
Smoothed reward
Episode Reward
Episode
160
180
200
220
240
260
280
300
0
200
400
600
800
1000
1200
1400


TD3 vs DDPG vs CAMO-TD3 - Reward Comparison
TD3
DDPG
CAMO-TD3
Episode Reward
Episode
300
280
260
240
220
200
180
160
0
200
400
600
800
1000
1200
1400